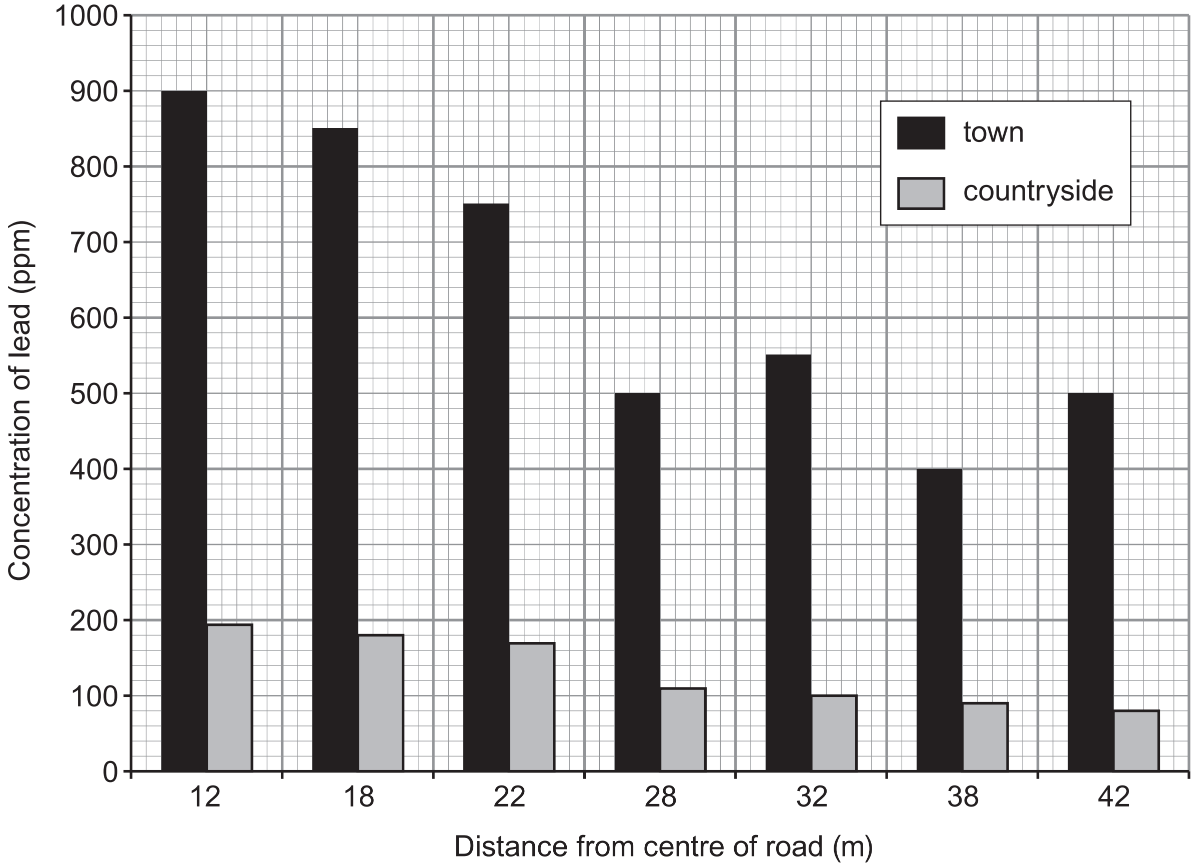
8. The most common understanding of the term 'heavy metal' is a metallic element which is toxic and has a high density, atomic number and relative atomic mass. The definitions used vary depending on the context. In metallurgy, for example, a heavy metal is defined on the basis of density, whereas in physics the distinguishing factor is atomic number. A chemist would likely be more concerned with chemical behaviour. More specific definitions have been published but none of these have been widely accepted.

Despite this lack of agreement, the term is widely used in science. Heavy metals are sometimes defined as metals with a density greater than 5 g / cm^3. They are often highly toxic or damaging to the environment. Chromium, arsenic, cadmium, mercury and lead have the greatest potential to cause harm on account of their extensive use.

Heavy metals are dangerous because they tend to bio-accumulate. Bio-accumulation occurs when the toxic chemical is taken into the body faster than it can be excreted. Lead can have an adverse impact on mental development in infants and children. Lead may also be a factor in behavioural problems. Heavy metal poisoning could result, for instance, from drinking-water contamination.

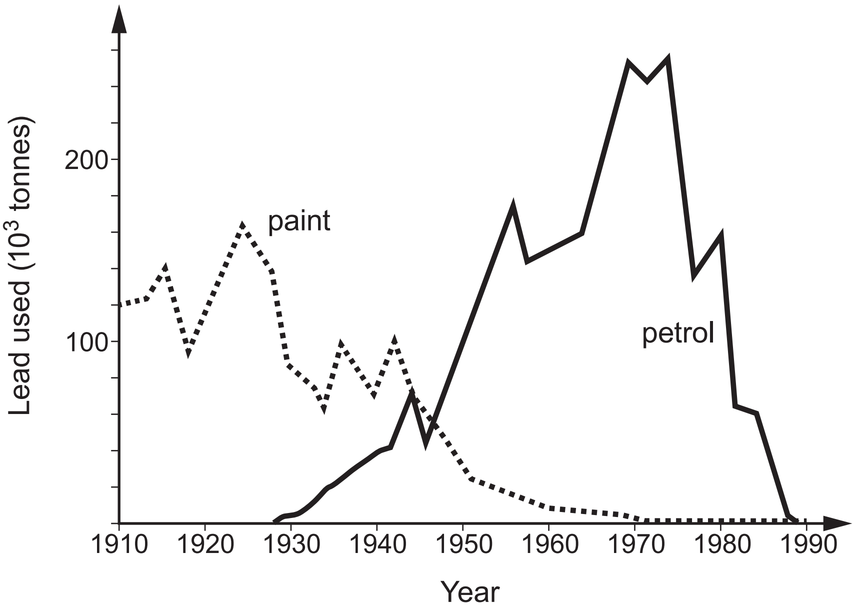
Lead is the most prevalent heavy metal contaminant. As a component of tetraethyl lead, $(CH_3CH_2)_4Pb$, it was used extensively in 'leaded petrol' from the 1930s-1970s. Although the use of leaded petrol has been phased out, soils next to roads can have high lead concentrations – see **Figure 1**. Lead-based paints were another early source of lead pollution but their use is now banned in the UK. **Figure 2** opposite shows how the amount of lead used in paint and petrol in the USA changed over the 20th century.

Figure 1



Examiner only

Figure 2



(a) (i) Put a tick (✓) in the box next to the statement which **best** describes why a single definition of a 'heavy metal' is not accepted by all scientists. [1]

- metallurgists, physicists and chemists do not trust each other ☐
- metallurgists, physicists and chemists are not concerned with the fact that heavy metals are toxic ☐
- metallurgists, physicists and chemists are concerned with different properties of heavy metals ☐
- metallurgists, physicists and chemists study different heavy metals ☐

(ii) Since the 1970s the use of lead water pipes has been prohibited across Europe. Explain why this is the case. [1]

....

....

Examiner only

8. Hydrogen peroxide solution, H_2O_2, is used in commercial stain removers.

A GCSE class investigated how effective four stain removers are at removing stains. Stain removers **A**, **B**, **C** and **D** contain different concentrations of hydrogen peroxide.

The students tested how effective each one is at removing an identical oil stain from four towels.

Their findings are outlined below.

Stain remover **A**	
Working temperature	= 50 °C
Cost per 100 cm^3	= 99p
Time to remove stain	= 40 min
Volume needed	= 20 cm^3

Stain remover **B**	
Working temperature	= 30 °C
Cost per 100 cm^3	= £1.99
Time to remove stain	= 40 min
Volume needed	= 10 cm^3

Stain remover **C**	
Working temperature	= 20 °C
Cost per 100 cm^3	= £2.49
Time to remove stain	= 20 min
Volume needed	= 5 cm^3

Stain remover **D**	
Working temperature	= 30 °C
Cost per 100 cm^3	= £1.49
Time to remove stain	= 30 min
Volume needed	= 10 cm^3

(a) In carrying out this investigation, which variables were kept the same in order to get valid results? Tick (✓) the correct answer. [1]

type of oil used, towel material and volume of hydrogen peroxide ☐

type of oil used, towel material and temperature of stain remover ☐

type of oil used and towel material ☐

type of oil used, towel material and cost of stain remover ☐

Examiner only

(b) Tick (✓) **all** of the statements which **could** explain why stain remover **A** has to be heated to 50 °C before it removes the stain. [1]

it is the cheapest stain remover ☐

it is heat resistant ☐

it has a low concentration of hydrogen peroxide ☐

it takes a long time to work ☐

(c) The students found that stain removers **B** and **D** used the same volume and worked best at the same temperature.

Assuming that they have the same hydrogen peroxide concentration, suggest a possible reason why **D** removes the stain more quickly than **B**. [1]

....

....

(d) One student went on to investigate the decomposition of hydrogen peroxide.

The equation for the reaction is as follows.

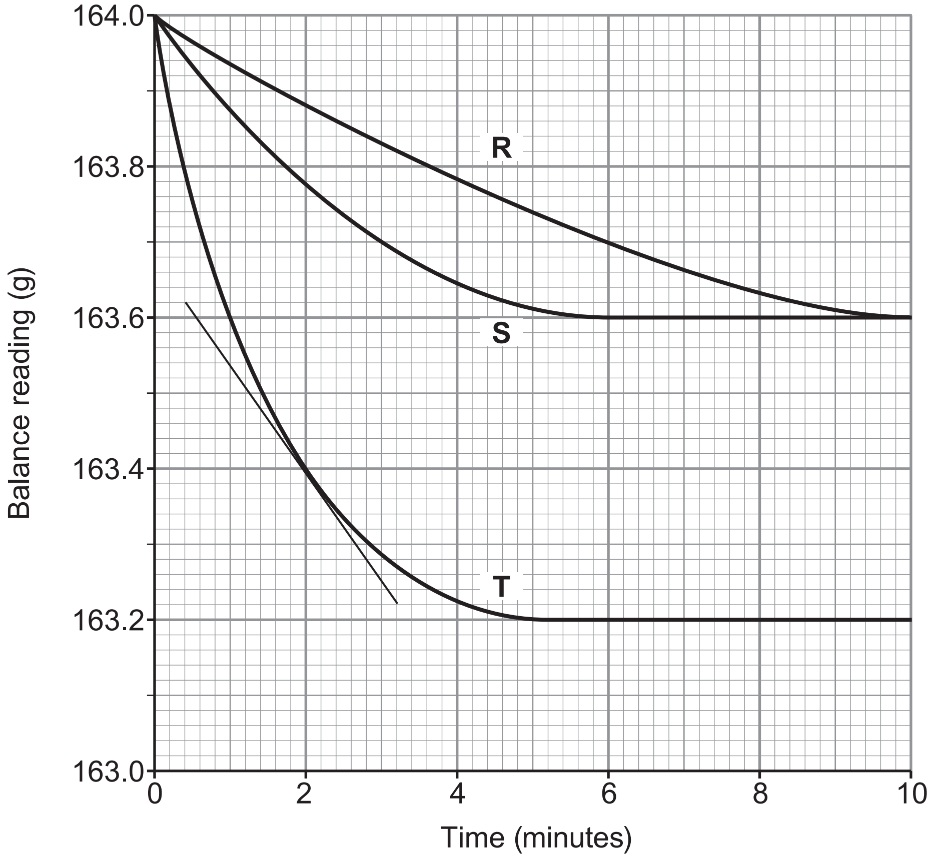
$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

The student investigated the effect of changing the concentration of the hydrogen peroxide solution on the rate of the reaction. She used manganese dioxide as a catalyst in each experiment.

This is the method she used.

- Pour 50 cm^3 of hydrogen peroxide solution of concentration **R** into a conical flask on a digital balance.
- Add 1 g of catalyst and place some cotton wool loosely in the neck of the flask. Record the balance reading and immediately start a stopwatch.
- Record the balance reading every minute until the mass no longer changes.
- Carry out the experiment twice more using hydrogen peroxide of different concentrations, **S** and **T**.

Her results are plotted on the grid below.



Examiner only

(i) Using the tangent shown on the graph, calculate the rate of reaction for concentration **T** at 2 minutes. Show your working. [2]

Rate at 2 minutes = ... g/minute

(ii) The initial rate for concentration **S** is half the initial rate for concentration **T**. Explain this difference in rate in terms of the particle theory. [3]

....

....

....

....

....

8

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2						Group					3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname		Centre Number	Candidate Number
First name(s)			0

GCSE

3430UB0-1

Z22-3430UB0-1

FRIDAY, 17 JUNE 2022 – AFTERNOON

SCIENCE (Double Award)

Unit 2 – CHEMISTRY 1
HIGHER TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**9**	
2.	**6**	
3.	**6**	
4.	**10**	
5.	**6**	
6.	**6**	
7.	**8**	
8.	**9**	
Total	**60**	

3430UB01
01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

Examiner only

8. The tables show the electronic structures of some Group 1 and Group 2 elements.

Group 1 metal	Electronic structure
sodium	2,8,1
potassium	2,8,8,1

Group 2 metal	Electronic structure
magnesium	2,8,2
calcium	2,8,8,2

(a) (i) Use the information to explain the trend in reactivity down Group 1. [2]

....

....

....

....

....

....

(ii) Use the information to explain the difference in reactivity of Group 1 and Group 2 elements. [2]

....

....

....

....

....

....

Examiner only

(b) Sodium reacts with water to produce sodium hydroxide and hydrogen. The equation for this reaction is shown.

$$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$$

Calculate the mass of sodium needed to produce 11.2 g of hydrogen gas. [3]

$A_r(Na) = 23$ $A_r(H) = 1$

Mass of sodium = ... g

(c) Group 2 metals react in a similar way with water as Group 1 metals.

The word equation for the reaction of calcium and water is shown.

calcium + water ⟶ calcium hydroxide + hydrogen

Write the balanced symbol equation for this reaction. [2]

....

9

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2					Group						3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
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133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

 (3430UB0-1)

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

S23-3430UB0-1

3430UB0-1

FRIDAY, 16 JUNE 2023 – MORNING

SCIENCE (Double Award)
Unit 2 – CHEMISTRY 1
HIGHER TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**8**	
2.	**7**	
3.	**4**	
4.	**6**	
5.	**8**	
6.	**9**	
7.	**9**	
8.	**9**	
Total	**60**	

3430UB01 01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

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INFORMATION FOR CANDIDATES

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Question **4** is a quality of extended response (QER) question where your writing skills will be assessed.

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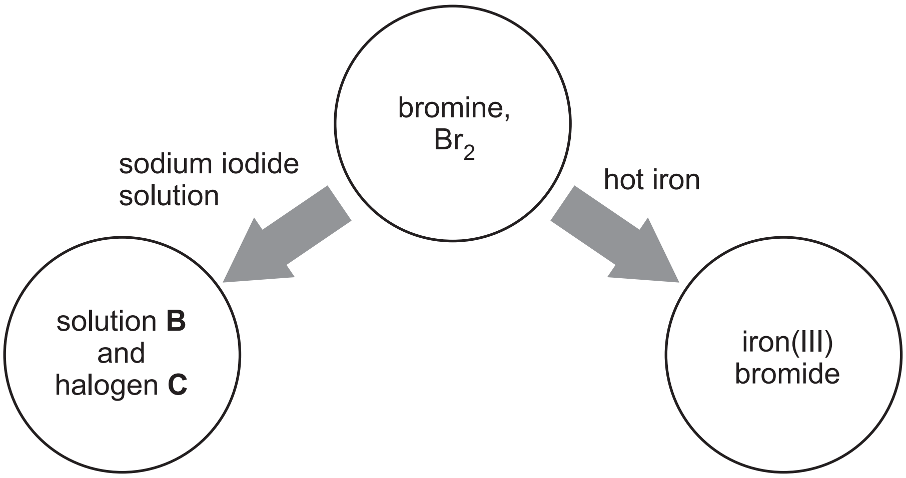
Examiner only

8. The diagram shows reactions of bromine, Br_2.



(a) (i) Give the names of solution **B** and halogen **C**. [2]

solution **B** ....

halogen **C** ....

(ii) Explain why bromine reacts with sodium iodide solution. [2]

....

....

....

(b) Write the symbol equation for the reaction between bromine, Br_2, and iron to form iron(III) bromide. [2]

....

(c) Silver nitrate solution was added to a solution of sodium iodide. Silver iodide was formed.

(i) State what you would expect to see. [1]

....

(ii) Give the **ionic** equation for the formation of silver iodide. [2]

....

END OF PAPER

9

FORMULAE FOR SOME COMMON IONS

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hydrogen	H^{+}	iodide	I^{-}
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iron(III)	Fe^{3+}	oxide	O^{2-}
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magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

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Group

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7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
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86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
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